INFORM Tanzania: Methodology & Data-Integrity Manual

Living document. Honest by design. This explains, end to end, how a raw measurement (a population count, a rainfall record, a flood event) becomes a 0–10 indicator, how indicators combine into the INFORM Risk, and how a Data-Entry edit travels the same path. Where a method is approximate, or a value looks too close to a neighbour, it is **flagged here openly** so we can improve it together, not hidden. Numbers below are real, taken from the current dataset (`src/data/tanzania-inform-risk.json`).

> **Superseded, 9 October 2026.** This manual describes the pipeline that built the dataset up to September 2026: computed drought and flood indices folded into the INFORM values with a “raise-only” floor, flood amplified by exposure, documented overlays and hand grades for the empty hazards, and the Kondoa district row quoted as the “official national INFORM value (4.1)”. The integrity assessment of October 2026 (`docs/assessment/INFORM_TZ_Integrity_Assessment_2026-10.md`) found that this changed 165 of 170 units of the INFORM SADC 2024 workbook and that the national figure was not national. Since then:
>
> - **The Core is the workbook, exactly** (`scripts/build-core.mjs`; reproduced by tests at 8,664/8,664 standardised values and 170/170 risks). Sections 1 and 5 below (the three aggregation rules) remain correct; the worked numbers now come from `docs/WORKED_EXAMPLE.md`, regenerated from the engine.
> - **The national figure is the INFORM aggregate of the 195 councils**: 3.8 (Hazard & Exposure 2.4, Vulnerability 3.8, Lack of Coping Capacity 6.2), labelled as not an official INFORM figure.
> - **Every Tanzanian computation is out of the Core.** Where kept, it is a documented sub-indicator of the **Advancement layer**, computed from its own inputs and shown on each profile beside the Core. There is no raise-only floor, no flood x exposure rule, no overlay and no hand grade anywhere.
> - **Councils carry their workbook unit’s values** until institutions enter council values through the Data portal (measured values, or 0-10 scores by the INFORM method with a citable source), reviewed by the owning institution’s chain; an approved council value applies to that council alone.
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> The current methodology is the Methodology page of the site (`/methodology`), the worked example and the assessment. Sections marked *legacy* below are kept as the record of the retired pipeline and must not be read as the current method.

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1. The risk journey (one picture)

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raw data ──standardize──▶ indicator 0–10 ──MEAN──▶ category 0–10
       └──────────────────────────────────────────────────┘
category 0–10 ──scaled GEOMEAN (Excel Box 6)──▶ dimension 0–10
Hazard&Exposure · Vulnerability · Lack-of-Coping ──∛(H×V×LCC)──▶ RISK 0–10 ──▶ class
```

Three rules, taken **exactly from the Tanzania INFORM Excel** (the INFORM SADC 2024 workbook, `data-source/tanzania-inform-risk.xlsx`, locked by `src/engine/risk/__tests__/workbook.golden.test.ts`):

Step	Method	Where
indicator → category	**arithmetic MEAN**	`AVERAGEIFS` in Excel
category → dimension	**INFORM scaled geometric mean**	Excel Box 6 (below)
dimension → risk	$\sqrt[3]{\textbf{H} \times \textbf{V} \times \textbf{LCC}}$ **(H × V × LCC)** (geometric mean)	INFORM Risk index

The same three rules run in the engine (`src/engine/risk/model.ts`) for the shipped Core and for every approved data update, so a workbook value and an institutional value behave identically.

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2. Standardization: turning a raw number into 0–10

Legacy in part (October 2026): the Core standardises every indicator by the workbook's own specification (denominator, outlier fence, logarithm, the indicator's reference range, direction), not by a min-max over Tanzania's districts; the population example below described a removed exposure indicator. See `docs/WORKED_EXAMPLE.md`, step 1.

Every indicator must sit on the same 0–10 scale ("higher = more risk") before it can be combined. We use **min-max** scaling across Tanzania's districts:

> **idx = (value − min) / (max − min) × 10**

For quantities that are highly skewed (a few huge cities, many small rural areas) we take the **log first** (log min-max), so the scale is not dominated by one outlier:

> **idx = ($\log_{10}$value − $\log_{10}$min) / ($\log_{10}$max − $\log_{10}$min) × 10**

Worked example: population → Exposure (your "3900" question)

Exposure is **population density** (people per km²), log-scaled because Dar (4,600/km²) dwarfs pastoral Longido (22/km²).

- Ilala (Dar): density 4,643/km² → $\log_{10}$ = 3.667 → **exposure 8.66**
- Kondoa: density 54/km² → $\log_{10}$ = 1.73 → **exposure 3.19**
- Longido: density 22/km² → $\log_{10}$ = 1.34 → **exposure 2.10**

So a district of (say) 3,900 people in a 1,300 km² ward = density 3.0/km² → near the bottom of the scale; the *same* 3,900 people in 5 km² = 780/km² → high. **Exposure is about concentration, not the raw headcount.** (Population comes from the NBS 2022 census, per council, summed to the district: see §7.)

Why min-max and not "rank"

An earlier draft ranked districts (1st, 2nd, …). That spreads everything evenly but **invents differences where the data is flat**: it once put random districts top of the drought list. Min-max keeps the *real* spacing: if two districts are genuinely similar, their scores stay close (and yes, sometimes they look “too close”, that is honest, see §8).

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3. Hazard indicators: what we compute, honestly

Legacy (retired October 2026): none of the computations in this section is in the Core. The drought composite was withdrawn (it scored wet highland councils 10); its members enter the Advancement layer separately. The flood and exposure rules, the floors and the overlays are removed.

3.1 Drought (computed, CHIRPS v3 + ERA5)

Drought is **slow** and hits communities through **failed growing seasons** (→ food insecurity, IPC/MUCHALI). SPI/SPEI *frequency* is ~16 % everywhere by construction, so it can NOT rank where drought is worse. Instead we blend four real signals, each min-maxed:

> **drought = minmax(0.30·aridity + 0.20·variability + 0.15·SPEI-depth + 0.35·season-failure) × 10**

- **aridity** = low mean annual rainfall (1991–2024 CHIRPS v3)
- **variability** = interannual rainfall CV
- **SPEI-depth** = mean magnitude of SPEI-12 during droughts (uses ERA5 temperature → Thornthwaite PET)
- **season-failure** = how often the main 4-month rainy season falls below 80 % of normal

Result (top): Longido 9.7, Simanjiro 9.1, Singida 8.7, Hanang 8.7, Kondoa 8.6: the real pastoral north + semi-arid centre. Bottom: Rungwe 0.5, Kyela 0.5, Mafia 1.5: the wet highlands/islands. *(Script: `scripts/compute-drought-spi-spei.py`.)*

The computed value is **floored at the documented baseline** (`drought = max(computed, documented)`): like flood, a known drought is never lowered, only raised where the climate evidence is stronger.

3.2 Heavy rainfall & flood (computed, CHIRPS daily + recorded events)

Two ingredients: 1. **Heavy-rain index**: count of days where the district-mean rainfall exceeds **50 mm** (heavy) and **100 mm** (very heavy), per year, 2015–2024 daily CHIRPS, min-maxed (very-heavy weighted ×2). Top: the Pemba/Zanzibar coast (Mkoani 14 heavy days/yr). 2. **Recorded floods**: a documented, *editable* event list (`data-source/flood_events.csv`) with sources (DesInventar / EM-DAT / PMO-DMD / news). `event index = min(10, 4 + 2 × #events)`.

The **physical flood hazard** = `max(heavy-rain index, event index)`, so a coastal cloudburst zone *or* a documented river-flood district both score high.

3.3 Flood = Hazard, amplified by Exposure (never lowered)

INFORM hazard is **Hazard × Exposure**, but exposure must only ever **raise** a flood score, never hide a flood-prone place. So:

> **hazardFlood = max(documented flood , heavy-rain index , recorded-event index) flood = max(hazardFlood , √(hazardFlood × exposure))**

Exposure lifts the score only where population *exceeds* the hazard (dense flood cities); a sparsely-populated but flood-prone district keeps its full documented hazard.

- **Ilala (Dar):** heavy-rain 4.0, 2 recorded floods (2011, 2018) → event index 8 → hazardFlood = **8**. Exposure 8.66 → √(8 × 8.66) = 8.3 → flood = max(8, 8.3) = **8.3**.
- **Pangani (coastal, flood-prone but sparse):** documented hazard 9.8, exposure 2.9 → √(9.8 × 2.9) = 5.3 → flood = max(9.8, 5.3) = **9.8: preserved.**

> ⚠️ An earlier draft multiplied flood *down* by exposure (Pangani 9.8 → 3.3, Korogwe 8 → 2.1: hiding known flood areas). That bug was caught by a before/after audit and fixed: **exposure now amplifies only; a documented flood hazard is never lowered.**

3.4 Other hazards: deepened one by one (all raise-only) *[legacy: removed from the Core]*

Each is floored at the documented baseline (`hazard_baseline.csv`): computed/overlay evidence can raise a hazard, never lower it. - **Storms & tropical cyclone**: coastal exposure overlay graded south→north, based on **Cyclone Hidaya 2024** (Mafia landfall, 18,862 affected), **Lindi cyclone 1952**, Zanzibar 1872. The INFORM baseline had ~0 here: outdated after Hidaya (first cyclone to hit Tanzania). - **Heatwave (new, computed)**: ERA5 temperature climatology (hot coast/lowlands → cool highlands), capped at 6.5 (Tanzania heat is moderate, not Sahel-extreme). `heatwave_era5.csv`. - **Lightning (new, overlay)**: Lake Victoria basin, one of Earth's top flash-density hotspots (NASA LIS/OTD; ~3,000–5,000 thunderstorm deaths/yr in the basin); shore districts 8–9. - **Volcano (new, overlay)**: Ol Doinyo Lengai (active, Lake Natron), Rungwe Volcanic Province, Kilimanjaro & Meru (dormant). - **Zoonoses / plants & pests (new, overlay)**: 2020 desert-locust invasion (north), fall armyworm (maize belts), livestock disease (pastoral), tsetse (western miombo). - **Earthquake, landslide, coastal hazards, wildfire, environmental degradation** keep their **documented / INFORM SADC** values (earthquake already follows the rift; the others are differentiated). *[Legacy plan: fire detections and the USGS event record are Advancement-layer members, never overlays on the Core.]* - **Vulnerability & coping** indicators are the next dimensions to deepen, so the Risk total will keep evolving as the model becomes more complete. That is expected and documented, not drift.

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4. Vulnerability & Coping: current sources

Legacy (retired October 2026): the Core's vulnerability and coping values are the workbook's (Global Data Lab, World Bank, WHO, UNHCR and others, per the workbook's Metadata sheet). The HBS, TDHS, THIS, UNHCR, IPC and census-facility rebuilds below

are out of the Core; institutions may re-enter them through the Data portal on the indicator's own definition.

- **Development & poverty**: NBS Household Budget Survey 2017/18 (poverty headcount); already differentiated by region (Kigoma/Geita/Kagera highest → Dar/Kilimanjaro lowest).
- **Children health & nutrition**: TDHS 2022 (under-5 stunting; Iringa/Njombe/Rukwa highest).
- **Health conditions (NEW)**: disease burden = **HIV** (THIS 2022-23: Njombe 11.4 % → Kigoma 1.7 %) + **malaria** (TDHS-MIS 2022: Tabora 23 %, Mtwara 20 %, Lake/West high; highlands ~0), min-max blended, raise-only. `health_burden.csv`.
- **Displaced people (NEW)**: **UNHCR 2024** refugees: Nyarugusu (Kasulu ~130 k) + Nduta (Kibondo ~75 k) → these were **0**, now **10 / 8** (burden-scaled). `displaced_refugees.csv`.
- **Livelihoods / food security (NEW)**: **IPC / MUCHALI** Nov 2023–Oct 2024: **Longido & Monduli in Phase 3 (Crisis)**, 16 more pastoral/semi-arid councils in Phase 2 (Stressed), applied raise-only. Only the **classified** councils get a value; the many baseline `0`s are **no-data** (INFORM SADC didn't classify them) and are left as-is, never fabricated. `food_security.csv`.
- **WASH**: derived from **real 2022-census water points & boreholes** (more resource → lower "lack of coping"), rank-scaled 2–8.
- **Access to health / education**: 2022 PHC facility counts.
- **DRR implementation**: EPRP / regional EOCC / drought Anticipatory-Action records (the districts you provided) lower the institutional "lack of coping".
- Indicators not yet localized keep the INFORM SADC 2024 baseline.

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5. Aggregation: the exact formulas

5.1 Indicator → Category: MEAN

`category = average of its indicators` (ignoring blanks). E.g. Rufiji Natural-hazards = mean(coastal 0, drought 2.9, earthquake 0.7, env. degradation 5.4, flood 9.0, landslide 5.2, storms 0, volcano 0, wildfire 4.5) = **3.08** (heatwave, lightning and zoonoses have no data and are excluded). *(Before the mean, each indicator group is itself the mean of its workbook indicators with data: see `docs/WORKED_EXAMPLE.md`, step 2.)*

5.2 Category → Dimension: INFORM scaled geometric mean (Excel Box 6)

> **dimension = (10 − GEOMEAN((10−c_1)/10·9 + 1 , (10−c_2)/10·9 + 1 , …)) / 9 × 10**

Because the scores are inverted before the geometric mean, the result is always **at or above** the plain average: a **high** category weighs more than a low one, so a serious problem in one category is not averaged away by a benign one.

Worked example: Rufiji Hazard: Natural 3.08, Human 2.50 (plain average 2.79). scaled: (10−3.08)/10·9+1 = 7.228 ; (10−2.50)/10·9+1 = 7.750. GEOMEAN = √(7.228 × 7.750) =

7.4845. dimension = ROUND((10 − 7.4845)/9 × 10, 1) = **2.8** (≥ 2.79). ✔ matches the engine.

> **Full end-to-end worked example** (real data, every indicator from raw value → standardisation → 0-10 → category → dimension → risk, plus how the four levels harmonise): see `docs/WORKED_EXAMPLE.md`, with the data tables in `docs/worked_example_indicators.csv` and `docs/worked_example_levels.csv`. It uses Rufiji (a documented flood district), its council Kibiti, and the Pwani region, regenerated from the live dataset by `scripts/build-worked-example.mjs`.

5.3 Dimension → Risk: cube root

> **Risk = ∛(Hazard × Vulnerability × LackOfCoping)**

Rufiji: ∛(2.8 × 3.8 × 6.0) = ∛63.84 = 3.995 → **4.0** (Medium). **Kondoa:** ∛(2.2 × 5.5 × 5.9) = ∛71.39 = **4.1** (Medium). ✔ (The in-app Methodology page recomputes this example live from the engine for any council.)

Then `Risk` is classed with the **Tanzania thresholds** (Very-Low < 2.5 < Low < 3.4 < Medium < 4.3 < High < 5.9 < Very-High).

5.4 Missing data: no-data never distorts the score

A blank indicator means **NO DATA** and is stored as `null`. Every aggregation (mean & scaled geomean) **excludes null entirely**: it is *not* read as 0: - `mean(5, 5, null) = 5`, not 3.33. **A no-data indicator can never pull a score up or down.** - `0` is different: it is a **real measured zero** (e.g. zero coastal-hazard exposure in landlocked Dodoma), and it **is** included. - An all-no-data category drops out cleanly (`null`), so the dimension is the scaled-geomean of the categories that actually have data.

In **Data Entry**, clearing a field sets it back to no-data (`null`): it removes data rather than zeroing it. Each district shows a **data-coverage %** (share of the 32 leaf indicators that have data) as a reliability cue, so a score built on sparse data is flagged for what it is. The ambiguous INFORM `economic` slot has no workbook data anywhere (170/170 null) and by design affects nothing. *(Locked by tests in `src/engine/risk/__tests__/`.)*

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6. Keying data in (Data Entry): same formulas, with provenance

Legacy (retired October 2026): the steps below describe the browser-only editor of the first version. Values now enter through the Data portal of the site: institutions submit measured values in natural units, or 0-10 scores by the INFORM method with a citable source, for the nation, a region or a council; the owning institution's reviewer approves; the engine recomputes every level.

1. Pick **role** (PMO/Admin apply directly; Sector officer submits for approval), region, district.

2. Edit a **dimension score** directly, or **Fill indicators** to set each 0–10 leaf, for **Hazard, Exposure, Vulnerability, Coping**.
3. **Exposure** has its own box; changing it instantly recomputes flood = √(hazard × exposure) and the Hazard score: the live preview uses the *same* mean → scaled-geomean → ∛ chain.
4. Choose the **Data source / authority** (NBS, TMA, MoW, MoH, MoA, PMO-DMD, DRR Coordinator, MUCHALI/IPC). Each edited indicator is **stamped** with that authority + date.
5. **Save / Submit → Apply.** The edit flows to the map, charts, tables. Hover the indicator and it shows **"✎ Updated by MoW · 2026-06"**; un-edited indicators show the registry source (e.g. **"📎 Source: TMA (+MoA, CHC) · CHIRPS v3 …"**).

Storage today is the browser (`localStorage`); the Supabase schema is ready for multi-user.

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7. Data sources (per indicator)

Legacy (retired October 2026): the Core's source for every indicator is the workbook's Metadata sheet (`src/data/inform-workbook-sources.json`), shown on the site beside each value. The table below lists the sources of the retired pipeline.

Indicator	Authority	Dataset	How standardized
Drought	TMA / MoA	CHIRPS v3 1991-2024 + ERA5 t2m	aridity+variability+SPEI-depth+season-fail, min-max
Flood	PMO-DMD / TMA / MoW	CHIRPS v3 daily + recorded floods	max(>50/100 mm count, events) × exposure
Exposure	NBS	2022 PHC population	density, **log** min-max
Poverty	NBS	HBS 2017/18	INFORM baseline
Stunting	MoH	TDHS 2022	INFORM baseline
Food security	MUCHALI / IPC (MoA)	IPC rounds	*to be entered*
WASH	MoW	2022 PHC water/boreholes	resource availability 2–8
Health/Education access	MoH / NBS	2022 PHC facilities	resource availability
DRR implementation	PMO-DMD / DRR Coordinator	EPRP/EOCC/AA records	investment overlay
Wildfire, earthquake, etc.	INFORM SADC 2024	INFORM	baseline (not yet recomputed)

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8. Honest limitations: the "to-improve" list (flag & discuss)

Legacy in part (October 2026): items 1 to 4b, 6, 8 and 9 describe the retired pipeline. The current limitations are the eleven items of the Methodology page (hazard aggregation rule, SADC classes, global sources, six empty groups, coherence, relative ranges, inheriting councils, shared units, Advancement status, precision, no reliability score).

1. **Drought exposure is agricultural, not urban.** We multiply *flood* by population exposure but **not drought**: multiplying drought by urban density would wrongly

demote pastoral communities (Longido). Better: cropland/livestock exposure. *Open.*

2. **Heavy-rain now uses CHIRPS v3 daily** (2015–2024, national download complete, ~96 % day coverage). Earlier drafts used v2.0 daily; v3 gives slightly lower extreme-day counts but the same spatial pattern (Pemba/Zanzibar/Mafia coast highest).
3. **Some values sit close together** (e.g. several semi-arid districts ~8.4–8.7 drought). That is real similarity, not precision: min-max preserves true spacing rather than forcing a spread. Don't over-read 0.1 differences.
4. **Dar es Salaam drought = 10 (documented artifact).** Ilala/Temeke carry drought 10 from the documented INFORM baseline: clearly too high for a coastal city. The precautionary floor keeps documented values rather than silently lowering them, so it persists (the *computed* value is ~6.6). **Flagged for correction via Data Entry** (MoA/TMA), which now overrides it. 4b. **Drought season in bimodal highlands (artefact).** The drought index scores the season it detects as the main rainy season. In some bimodal highland councils it selects the erratic short rains (Oct–Jan) instead of the reliable long rains: **Lushoto computes to 10.0** (season-failure 0.35, SPEI 1.56), which is implausible for the wet Usambara highlands. *Open*: score both seasons in bimodal areas (or the livelihood-relevant one) and validate against MUCHALI/IPC history; meanwhile read very high drought in wet highland councils with caution.
5. **Administrative structure: the model now speaks the real NBS-2022 structure by default.** The explorer, the ranked table, the charts and Data Entry all open on the **195 councils (LGAs)**, aggregating to the **31 regions**, with the **INFORM aggregate of the 195 councils (3.8, not an official INFORM figure)** as the national figure *[corrected October 2026: the "official 4.1" quoted here before was Kondoa's row]*. This is a *consistent scaling of the same source*: the model was built from the INFORM Tanzania **country-model workbook** (170 district-level units, the **reference**), and the 195 NBS-2022 councils are derived from it. **195/195 councils carry data**: 175 take the values of their workbook unit directly; **20 councils created after the workbook's boundaries** (Kibiti←Rufiji, Ubungo←Kinondoni, Kigamboni←Temeke, Mlimba/Ifakara←Kilombero…) **inherit their parent's values, flagged**, until council values are approved (**never fabricated**). Because the data is 170-resolution, **25 workbook units back two councils each**. A council carries its unit's values for all three dimensions; an **approved council value (measured or scored) applies to that council alone**, and the Data portal shows which councils each change reaches. Council risk is computed live as $\sqrt[3]{(H \times V \times LCC)}$, so approved values reach the map, regions and profiles immediately. **Regions aggregate from their councils the INFORM way** (indicator means → groups → categories → scaled geomean → cube root; facilities summed over distinct workbook units, so no double counting). The **170 workbook units remain available as a labelled reference level**. Official NBS 2022: 31 regions · 150 districts · 195 councils · 4,344 wards. Reconciliation: `scripts/export-councils.py` → `tanzania-councils.json` + `council_reconciliation.csv`.
6. **Wildfire, earthquake, lightning, storms** are the workbook's values in the Core (earthquake is the GEM exposure index). Fire detections, the USGS event record and LIS/OTD lightning density are Advancement-layer members, present or planned, never Core sources.
7. **Standardization uses the workbook's own reference ranges** (17 fixed pairs and 36 data ranges over the 170 units, frozen at the 2024 release), not INFORM Global's. Subnational models are not comparable with INFORM Global; never set the two side by side as if they were.

8. **Precautionary by design** *[legacy: the floor was removed in October 2026; the Core is the workbook and the Advancement layer is shown beside it]*. Flood and drought were *floored* at the documented baseline: computed evidence could **raise** a hazard but never **lower** it. This errs toward flagging risk (safer for planning) at the cost of keeping a few possibly-overstated documented values (see #4). A before/after audit confirms **0 districts** had flood or risk-class lowered.
9. **Risk evolves as the model is completed, by design** *[legacy: these fills were hand grades and overlays inside the Core and were removed in October 2026; the six empty groups stay empty in the Core until filled in the Advancement layer or by institutions]*. Populating previously-blank hazards (storms, heatwave, lightning, volcano, zoonoses, earthquake-refine, wildfire-refine, plus the technological hazards road-accidents & hazmat) raised High-risk districts from **18 → 43**, all raise-only (a value-audit confirms **0 districts below their documented baseline**). The coast (cyclone + heat), the Lake-Victoria zone (lightning + flood) and the western rift (earthquake) are the main risers. **Technological hazards (vehicleAccidents, hazmat) are kept deliberately moderate**: human hazards carry 50 % of the Hazard dimension, so road accidents should not dominate *disaster* risk. Deepening **vulnerability & coping** next will shift it again, toward a more complete picture, not drift. Every change is traceable to a sourced indicator.

Everything in this list is editable in Data Entry and improvable in code: that is the point of writing it down.

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9. Risk over time (snapshots)

Risk is reported **“as of”** a date (`metadata.asOf`, currently the baseline). The plan to turn this into a time series, with dated snapshots, detection of **emerging** (risk ↑) and **improving** (risk ↓) areas and a Δrisk map, is in **RISK_OVER_TIME_PLAN.md**. Foundation already in place: every Data-Entry edit is timestamped with its authority, the CHIRPS inputs are inherently temporal (1991–2024), and the trend-chart components exist, so future seasonal refreshes can be compared to spot communities whose risk is rising before it becomes a disaster.